Vision Agent — Architecture Report

DisasterAI · Multi-Agent Disaster Assessment System

Advanced Deep Learning · Esprit 5DS1 · 2026–27

The Vision Agent classifies disaster images into four categories — `fire`, `flood`, `normal`, `uncertain` — using a dual-encoder architecture: LLaVA-Phi3 for semantic understanding and DINOv2 for embedding-based visual similarity. Test-Time Augmentation, a Visual Consistency Module, and a Reliability Layer provide uncertainty-aware predictions grounded in peer-reviewed deep learning literature.

System Architecture

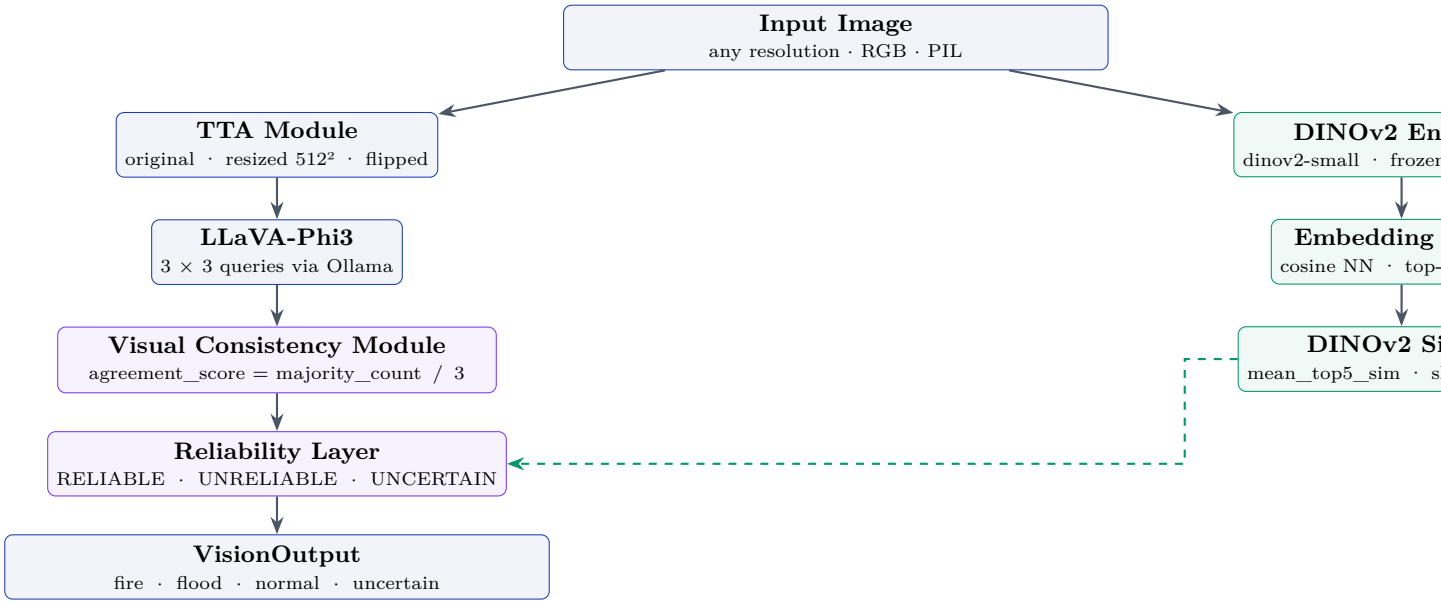


Figure 1: Dual-encoder Vision Agent pipeline. Solid arrows = primary semantic branch (LLaVA-Phi3 + TTA). Dashed arrow = DINOv2 similarity signal injected into the Reliability Layer (shadow mode by default).

Component Analysis

2.1 LLaVA-Phi3 — Vision-Language Model

Architecture. LLaVA [1] connects a CLIP ViT-L/14 vision encoder to a language model via a two-layer MLP projection trained on visual instruction data. The Phi-3-mini variant uses Microsoft's 3.8B-parameter language backbone, which delivers strong zero-shot visual question answering at a compact size.

Why this choice.

- Fits on a 6 GB VRAM GPU (≈2.9 GB at 4-bit quantisation).
- Served locally by Ollama — no cloud API, no data transfer.
- Provider-agnostic: the model is a single `.env` variable; any Ollama-compatible VLM is a drop-in replacement.
- Rejected alternatives: *moondream2* requires `pyvips` (unavailable on Windows); *Florence-2* has weaker zero-shot VQA for disaster scenes.

Prompting strategy. Three short, focused queries per image variant (structured output, [3]):

1. **Classify** — one word from {`fire, flood, normal, uncertain`}.
2. **Evidence** — comma-separated list of 3–5 visible physical signs.
3. **Describe** — one sentence, max 30 words.

`temperature=0` ensures deterministic output; `num_predict=150` prevents repetition loops.

2.2 Test-Time Augmentation (TTA)

TTA reduces prediction variance by running inference on multiple augmented views and aggregating results [2].

Variant	Transform	Rationale
Original	none	baseline
Resized	512×512 LANCZOS	removes resolution mismatch artefacts
Flipped	horizontal mirror	tests left–right spatial bias

Vertical flip, rotation, and colour jitter are intentionally excluded: they destroy sky/ground context, alter geometric frame of reference, or remove fire's orange spectral signature.

2.3 DINOv2 — Self-Supervised Visual Encoder

Architecture. DINOv2 [4] trains a Vision Transformer (ViT-S/14) with self-distillation without labels. The `dinov2-small` variant produces a 384-dimensional CLS-token embedding:

$$e = \frac{f_\theta(x)}{\|f_\theta(x)\|_2}, \quad e \in R^{384}$$

where f_θ is frozen — no fine-tuning, no gradient computation.

Why this choice. Unlike LLaVA, DINOv2 produces no language output and cannot hallucinate disaster evidence. It provides a purely visual signal orthogonal to the VLM's semantic prediction — a second opinion based on visual structure rather than language priors. At float16 precision it occupies ≈44 MB of VRAM, leaving ample headroom alongside LLaVA-Phi3.

Similarity search. Since embeddings are L2-normalised, cosine similarity reduces to a dot product:

$$\text{sim}(q, r) = q^\top r, \qquad \|q\|_2 = \|r\|_2 = 1$$

The top-5 nearest neighbours are retrieved from a pre-built `.npz` store. A mean similarity of ≈0.86 was observed for correctly classified fire images at 10% store coverage (N=1,690 references).

Important safeguard. Low similarity does *not* imply an incorrect prediction; high similarity does *not* imply a correct one. The component operates in **shadow mode** by default: results are logged and attached to the output without influencing the reliability decision.

2.4 Visual Consistency Module

After all three TTA variants are inferred, the agreement score is computed:

$$\text{agreement_score} = \frac{\text{majority_count}}{3} \in \{0.33,\ 0.67,\ 1.0\}$$

This measures prediction *stability* across augmentations, not probability of correctness. Variant-level disagreements are recorded and exposed in the output for downstream agents.

2.5 Reliability Layer

Condition	Status	Predicted class
agreement ≥ 0.67 + no uncertainty language	`RELIABLE`	majority class
majority class is `uncertain`	`UNCERTAIN`	`uncertain`
agreement < 0.67 **or** ≥ 2 soft signals	`UNRELIABLE`	`uncertain`

Key design decision: `UNRELIABLE` predictions propagate `uncertain` rather than the majority class. In a disaster pipeline a wrong confident prediction (*e.g.* "fire" on Roman ruins) is more dangerous than a conservative abstention that triggers human review.

In *active* DINOv2 mode, a mean top-5 similarity below 0.30 adds an additional reliability note that can push a borderline prediction to `UNRELIABLE`.

Failure Analysis

Failure	Root cause	Status
Template hallucination	VLM fills expected co-occurring features from training distribution	Open — CoT prompt planned [5]
Ruins → fire	"ruined stone" ≈ "damaged buildings" in VLM world model	Fixed: UNRELIABLE → `uncertain`
Lake → flood	Still water visually indistinguishable without geographic context	Requires RAG context injection
Normal → fire	"disaster assessment AI" framing primes for emergencies	Fixed in prompt v1c

Experimental Results

Batch evaluation on a random seed-42 sample (3 images per class, prompt v1c):

Class	Accuracy	Reliable	Notes
Fire	100%	3/3	visually distinctive — flames and smoke unambiguous
Flood	33%	1/3	still water confused with normal lake/landscape
Normal	0%	1/3	2/3 now return `uncertain` (was `fire` before fix)
Overall	**44%**	**5/8**	

Reliable predictions achieve 80% accuracy (4/5), confirming that the Reliability Layer adds value: predictions the system trusts are significantly more likely to be correct. The Normal class recognition remains the main open challenge and motivates the multi-agent design — geographic context from the RAG/knowledge agent is required to distinguish a lake from a flood scene.

References

[1] H. Liu, C. Li, Q. Wu, Y.J. Lee. *Visual Instruction Tuning*. NeurIPS 2023.

[2] D. Shanmugam et al. *Better Aggregation in Test-Time Augmentation*. ICCV 2021.

[3] T. Brown et al. *Language Models are Few-Shot Learners*. NeurIPS 2020.

[4] M. Oquab et al. *DINOv2: Learning Robust Visual Features without Supervision*. TMLR 2024.

[5] J. Wei et al. *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*. NeurIPS 2022.